

Unit-Free Multiplicative Linear Logic is Idempotent Complete

Rémi Di Guardia

Université Paris Cité, Inria, CNRS, IRIF, Paris, France

remi.di.guardia@ens-lyon.org

Abstract

This paper shows that unit-free multiplicative linear logic is idempotent complete: every idempotent splits, that is every idempotent can be written as a section composed with a retraction. Interestingly, this decomposition can be directly obtained by applying on the given idempotent an algorithm for “proof-relevant” interpolation – whereas “non-proof-relevant” interpolations cannot yield anything.

1 Introduction

We consider the sequent calculus of unit-free multiplicative linear logic [Gir87]. Proofs of this system are often considered up to *cut-elimination* and *axiom-expansion*, which is the usual quotient, denoted $=_{\beta\eta}$, to obtain a category whose objects are formulas and whose morphisms are equivalence classes of proofs, see *e.g.* [See89]. Indeed, a proof π of $\vdash A^\perp, B$ and a proof ϕ of $\vdash B^\perp, C$ can be composed by a *cut*-rule into the proof

$$\frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\phi}{\vdash B^\perp, C}}{\vdash A^\perp, C} (cut). \text{ Moreover, there always is a proof } \overline{\vdash A^\perp, A}^{(ax)} \text{ of } \vdash A^\perp, A.$$

Quotienting up to cut-elimination and axiom-expansion then yields the associativity of composition by the *cut*-rule, and that the *ax*-rule is neutral for it.

This paper studies *idempotents* in this framework, and their links with *section-retraction pairs*. An **idempotent** is a proof τ of $\vdash A^\perp, A$ such that:

$$\frac{\frac{\tau}{\vdash A^\perp, A} \quad \frac{\tau}{\vdash A^\perp, A}}{\vdash A^\perp, A} (cut) =_{\beta\eta} \tau \quad (1)$$

A **section-retraction pair** is a pair of proofs π and ϕ of respective conclusions $\vdash A^\perp, B$ and $\vdash B^\perp, A$ such that:

$$\frac{\frac{\phi}{\vdash B^\perp, A} \quad \frac{\pi}{\vdash A^\perp, B}}{\vdash B^\perp, B} (cut) =_{\beta\eta} \overline{\vdash B^\perp, B}^{(ax)} \quad (2)$$

if (π, ϕ) is such a pair then $\tau \stackrel{\text{def}}{=} \frac{\pi \quad \phi}{\vdash A^\perp, A} \text{ (cut)}$ is an idempotent. Indeed:

Such an idempotent made from a section-retraction pair is called a **split idempotent**.

i.e. any idempotent τ can be decomposed as $\tau =_{\beta\eta} \frac{\pi \quad \phi}{\vdash A^\perp, A} \vdash B^\perp, A \text{ }_{(cut)}$ with (π, ϕ)

Back to linear logic, how to obtain the section-retraction pair (π, ϕ) from τ is easy: it suffices to apply the algorithm from Alexis Saurin’s proof-relevant interpolation [Sau25] to τ . More exactly, we use a *slightly stronger theorem that the one stated in [Sau25] but whose proof is the one in [Sau25]*.

Theorem 1 ([Sau25, Theorem 5 reformulated]). *Let τ be a cut-free proof of the sequent $\vdash A^\perp, B$, with A and B two arbitrary formulas. There exists a formula C and cut-free proofs π and ϕ of respectively $\vdash A^\perp, C$ and $\vdash C^\perp, B$ such that $\frac{\pi \quad \phi}{\vdash A^\perp, B} \text{ (cut)}$ reduces by cut-elimination to τ and no ax -rule in π (respectively ϕ) uses two sub-formula occurrences of C (respectively of $C^\perp$). Plus, if τ is ax -expanded, then so are π and ϕ .*

The strengthening compared to the statement in [Sau25] is the last part in italic about ax -rules.¹ Please observe that such a *proof-relevant* interpolation is strictly stronger than Lyndon’s interpolation, hence than Craig’s interpolation too. Moreover, it is crucial here: the usual Craig’s and Lyndon’s interpolations are useless for our purposes, since an idempotent τ is a proof of $\vdash A^\perp, A$ so that looking at formulas cannot yield anything interesting; we really need to look at the proof τ itself. Thanks to Theorem 1, to prove an idempotent τ is split it suffices to prove that the two resulting interpolation proofs π and ϕ form a section-retraction pair. This is what we show in this paper.

Since we work in unit-free multiplicative linear logic, it would make sense to use not the sequent calculus syntax but the proof-net one [Gir96]. Indeed, this syntax is more canonical, making problems “up to $=_{\beta\eta}$ ” easier to consider. Furthermore, Alexis Saurin extended Theorem 1 to proof-nets working with Guido Fiorillo and Daniel Osorio Valencia [FOS25].² Nonetheless, we chose not to do so since we hope to extend our result beyond unit-free multiplicative linear logic, adding units, additive connectives or exponential connectives, in which cases proof-nets are not as canonical. Still, since intuitions might be clearer at the level of proof-nets, the key points in our argumentation will be illustrated by proof-nets – and for the reader familiar with proof-nets it should be most easy to translate our demonstration from sequent calculus to the proof-net syntax.

2 Unit-Free Multiplicative Linear Logic

We present here unit-free multiplicative linear logic in a standard way as a *unilateral* sequent calculus [Gir87]. **Formulas** are defined by the following grammar, with X an atom in some countable set:

$$A, B ::= X \mid X^\perp \mid A \wp B \mid A \otimes B$$

Formulas of the kind X or $X^\perp$ are called **atomic**. **Orthogonality** $(\cdot)^\perp$ (*a.k.a.* negation, duality) is extended as an involution to arbitrary formulas using the De Morgan’s laws:

$$(X^\perp)^\perp \stackrel{\text{def}}{=} X \quad (A \wp B)^\perp \stackrel{\text{def}}{=} A^\perp \otimes B^\perp \quad (A \otimes B)^\perp \stackrel{\text{def}}{=} A^\perp \wp B^\perp$$

¹Such a restriction on ax -rules was already considered when studying section-retraction pairs in [Di24] and called being *half-bipartite*.

²In the proof-net syntax, that the part we added to Theorem 1 still holds can also be seen by applying the algorithm expanding *cut* in [Di+26, Appendix E].

Rules of the logic are the following, where A and B stand for arbitrary formulas, and Γ and Δ for sets of (occurrences of) formulas:

$$\frac{}{\vdash A^\perp, A} \text{ (ax)} \quad \frac{\vdash A, \Gamma \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} \text{ (cut)} \quad \frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma} \text{ (\wp)} \quad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} \text{ (\otimes)}$$

This sequent calculus is equipped with a **cut-elimination** and an **axiom-expansion** procedures (not recalled here) that are weakly normalising [LL 22], and whose normal forms are respectively *cut-free* proofs (with no *cut*-rule) and *ax-expanded* proofs (with *ax*-rules only on atomic formulas). Plus, cut-elimination preserves being an *ax-expanded* proof. **Equality up to cut-elimination and axiom-expansion** is denoted by $=_{\beta\eta}$.

We recall that **idempotents**, **split idempotents** and **section-retraction pairs** are defined in the introduction (Section 1).

3 Around *ax*-rules and Occurrences

We do not define properly occurrences, but use the following convention. We use indexes to distinguish different occurrences of a same atom, and when X_i is an occurrence of X in some formula C , then $X_i^\perp$ is the *dual* occurrence of X_i in $C^\perp$ as identified by the De Morgan's laws defining $(\cdot)^\perp$.

We will need in our demonstration the following known or easy results about *ax*-rules and occurrences of atoms.

Fact 2. *Let π be an *ax-expanded* proof of the sequent $\vdash \Gamma$. Then, each occurrence of an atomic formula from $\vdash \Gamma$ is in the conclusion of a unique *ax*-rule of π .*

Proof. By induction on π . □

Fact 3. *Let π be a proof of conclusion $\vdash \Gamma$ that contains the *ax*-rules $\frac{}{\vdash X_1^\perp, X_2} \text{ (ax)}$, $\frac{}{\vdash X_2^\perp, X_3} \text{ (ax)}$, $\dots$, $\frac{}{\vdash X_{n-1}^\perp, X_n} \text{ (ax)}$ where $X_1^\perp$ and X_n occur in Γ , and $X_i^\perp$ and X_i are dual occurrences of a formula introduced by a *cut*-rule in π for all $1 < i < n$. There is an *ax*-rule $\frac{}{\vdash X_1^\perp, X_n} \text{ (ax)}$ in every cut-free ϕ such that π reduces by cut-elimination to ϕ .*

Proof. By induction on the reduction sequence $\pi \rightarrow^* \phi$. The only cut-elimination step modifying some *ax*-rule is the *ax* key case:

$$\frac{\frac{\tau}{\vdash A, \Delta} \quad \frac{}{\vdash A^\perp, A} \text{ (ax)}}{\vdash A, \Delta} \text{ (cut)} \longrightarrow \frac{\tau}{\vdash A, \Delta}$$

This step may delete a $\frac{}{\vdash X_{i-1}^\perp, X_i} \text{ (ax)}$ and the occurrence X_i (respectively $X_{i-1}^\perp$), so that the *ax*-rule using the occurrence $X_i^\perp$ (respectively X_{i-1}) now use $X_{i-1}^\perp$ (respectively X_i) instead. Since in the cut-free ϕ only occurrences of Γ are present, we can only end with $\frac{}{\vdash X_1^\perp, X_n} \text{ (ax)}$. □

Lemma 4. *Let π and ϕ be cut-free and ax -expanded proofs of the same sequent. Then $\pi =_{\beta\eta} \phi$ if and only if the ax -rules of π and ϕ are exactly on the same occurrences.*

Proof. This is a corollary of [Di 24, Theorem 2.50], [HG16, Theorem 1] and the obvious fact that π and ϕ have the same proof-net if and only if their ax -rules are exactly on the same occurrences. $\square$

4 Splitting an Idempotent

In all this section, we assume given an idempotent τ of conclusion sequent $\vdash A^\perp, A$. Without any loss of generality, we assume that τ is cut-free and ax -expanded. By applying Theorem 1 on τ , we obtain a formula B and two cut-free and ax -expanded proofs π and ϕ respectively of $\vdash A^\perp, B$ and of $\vdash B^\perp, A$ such that:

- (a) $\frac{\pi \quad \phi}{\vdash A^\perp, B \quad \vdash B^\perp, A} \text{ (cut) } \text{ reduces by cut-elimination to } \tau;$
- (b) no ax -rule in π (respectively ϕ) uses two atomic formulas occurring in B (respectively in $B^\perp$).

We claim that π and ϕ form a section-retraction pair. More precisely, we will prove that

$\frac{\phi \quad \pi}{\vdash B^\perp, A \quad \vdash A^\perp, B} \text{ (cut) } \text{ is equal, up to cut-elimination, to the full axiom-expansion of } \frac{\vdash B^\perp, B}{\vdash B^\perp, B} \text{ (ax)}.$ ³ It is hence natural to consider an arbitrary normal – *i.e.* cut-free – form μ of $\frac{\phi \quad \pi}{\vdash B^\perp, A \quad \vdash A^\perp, B} \text{ (cut)}$. To deduce our claim, it suffices to prove the following two results about μ .

Lemma 5. *In each ax -rule $\frac{}{\vdash X^\perp, X} \text{ (ax)}$ of μ (for any atom X), one of $X^\perp$ and X occurs in $B^\perp$ and the other occurs in B .*

Lemma 6. *Consider in μ an ax -rule $\frac{}{\vdash X^\perp, X} \text{ (ax)}$ where one of $X^\perp$ and X occurs in $B^\perp$ and the other in B . Then $X^\perp$ and X are dual occurrences in $B^\perp$ and B .*

Once these two results on μ are proved, we are done. Indeed, together they imply that each and every ax -rule of μ is between one atomic formula of B and its dual occurrence in $B^\perp$. The only proof respecting such a property is an axiom-expansion of $\frac{}{\vdash B^\perp, B} \text{ (ax)}$ – or proofs that are equal to it up to cut-elimination (Lemma 4).

³As an example of simplification when using proof-nets, in that setting one can replace “is equal up to cut-elimination” with “reduces by cut-elimination”.

We now proceed to prove these two intermediate results, then conclude the demonstration of our theorem. Observe that:

$$\begin{aligned}
& \frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\mu}{\vdash B^\perp, B}}{\vdash A^\perp, B} (cut) \quad \frac{\phi}{\vdash B^\perp, A} \\
& \frac{\vdash A^\perp, B \quad \vdash B^\perp, A}{\vdash A^\perp, A} (cut) \\
& =_{\beta\eta} \frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\frac{\phi}{\vdash B^\perp, A} \quad \frac{\pi}{\vdash A^\perp, B}}{\vdash B^\perp, B} (cut)}{\vdash A^\perp, B} (cut) \quad \frac{\phi}{\vdash B^\perp, A} \quad \text{by definition of } \mu \\
& \frac{\vdash A^\perp, B \quad \vdash B^\perp, A}{\vdash A^\perp, A} (cut) \quad \frac{\vdash A^\perp, B \quad \vdash B^\perp, A}{\vdash A^\perp, A} (cut) \quad \text{by associativity of } cut \\
& =_{\beta\eta} \frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A} (cut) \quad \frac{\pi}{\vdash A^\perp, B} \quad \frac{\phi}{\vdash B^\perp, A} \\
& =_{\beta\eta} \frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A} (cut) =_{\beta\eta} \tau \quad \text{by idempotence of } \tau \text{ and Item (a)}
\end{aligned}$$

Hence the following property:

$$\begin{aligned}
& \text{(c) the normal forms of } \frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\mu}{\vdash B^\perp, B}}{\vdash A^\perp, B} (cut) \quad \frac{\phi}{\vdash B^\perp, A} \quad \text{and those of} \\
& \frac{\vdash A^\perp, B \quad \vdash B^\perp, A}{\vdash A^\perp, A} (cut) \quad \text{are all equal to } \tau \text{ up to } =_{\beta\eta}.
\end{aligned}$$

We will use Item (c) to prove Lemmas 5 and 6, whose demonstrations are illustrated in a “proof-net way” (*c.f.* Figures 1 and 2) where triangles represent formula trees and **dashed** or **dotted** edges stand for *ax*-rules between two occurrences.

Proof of Lemma 5. We proceed by *reductio ad absurdum*. Assume μ contains an *ax*-rule $\frac{}{\vdash X_1^\perp, X_2^\perp} (ax)$ with X_1 and $X_2^\perp$ both occurring in B – a similar argumentation handles the symmetric case where those are occurring in $B^\perp$. We use Item (c) and Lemma 4 to find in τ incompatible *ax*-rules. See Figure 1 for a “proof-net illustration”.

We assumed in μ an *ax*-rule $\frac{}{\vdash X_1^\perp, X_2^\perp} (ax)$ with X_1 and $X_2^\perp$ both occurring in B . Consider the occurrences X_1 and $X_2^\perp$ of $B^\perp$: by Fact 2, there are in ϕ *ax*-rules $\frac{}{\vdash X_1, X_3^\perp} (ax)$ and $\frac{}{\vdash X_2^\perp, X_4} (ax)$, with $X_3^\perp$ and X_4 both occurring in A by Item (b) – see the upper half

of Figure 1. Therefore, in any normal form of $\frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\mu}{\vdash B^\perp, B}}{\vdash A^\perp, B} (cut) \quad \frac{\phi}{\vdash B^\perp, A} (cut)}{\vdash A^\perp, A} (cut)$

there is an ax -rule $\frac{}{\vdash X_3^\perp, X_4} (ax)$ by Fact 3, hence also in τ by Item (c) and Lemma 4.

Consider now $\frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A} (cut)$. Recall there are in ϕ ax -rules $\frac{}{\vdash X_1^\perp, X_3^\perp} (ax)$ and $\frac{}{\vdash X_2^\perp, X_4^\perp} (ax)$ with $X_3^\perp$ and $X_4^\perp$ both occurring in A . Looking at the occurrences

$X_1^\perp$ and $X_2^\perp$ of B , by Fact 2 there are in π ax -rules $\frac{}{\vdash X_1^\perp, X_5^\perp} (ax)$ and $\frac{}{\vdash X_2^\perp, X_6^\perp} (ax)$ with $X_5^\perp$ and $X_6^\perp$ both occurring in $A^\perp$ by Item (b) – see the lower half of Figure 1.

Therefore, in the normal form τ of $\frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A} (cut)$ (Item (a)) there are ax -rules $\frac{}{\vdash X_3^\perp, X_5^\perp} (ax)$ and $\frac{}{\vdash X_4^\perp, X_6^\perp} (ax)$ by Fact 3.

We now have reached a contradiction. In τ there is both an ax -rule $\frac{}{\vdash X_3^\perp, X_4^\perp} (ax)$ and an ax -rule $\frac{}{\vdash X_3^\perp, X_5^\perp} (ax)$, with $X_4^\perp$ occurring in A but $X_5^\perp$ occurring in $A^\perp$ so that X_4 and X_5 are not the same occurrence. This is impossible through Fact 2. $\square$

Proof of Lemma 6. Consider in μ an arbitrary ax -rule $\frac{}{\vdash X_1^\perp, X_2^\perp} (ax)$ with $X_1^\perp$ an occurrence from $B^\perp$ and $X_2^\perp$ an occurrence from B – a similar argumentation handles the symmetric case where $X_1^\perp$ is an occurrence from B and $X_2^\perp$ is an occurrence from $B^\perp$. Let us prove that $X_1^\perp$ and $X_2^\perp$ refer to the same occurrence. As in the proof of Lemma 5, we find some ax -rules using Fact 2 and Item (b) – see Figure 2 for a “proof-net illustration”:

- in π there is $\frac{}{\vdash X_3^\perp, X_1^\perp} (ax)$ where $X_3^\perp$ occurs in $A^\perp$;
- in ϕ there is $\frac{}{\vdash X_1^\perp, X_4^\perp} (ax)$ where $X_4^\perp$ occurs in A ;
- in ϕ there is $\frac{}{\vdash X_2^\perp, X_5^\perp} (ax)$ where $X_5^\perp$ occurs in A .

Let us first consider the proof $\frac{\frac{\pi}{\vdash A^\perp, B} \quad \frac{\frac{\mu}{\vdash B^\perp, B} \quad \frac{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, B} (cut)}{\vdash A^\perp, A} (cut)$. Thanks to Fact 3

and following the chain of ax -rules $\frac{}{\vdash X_3^\perp, X_1^\perp} (ax)$ (in π), $\frac{}{\vdash X_1^\perp, X_2^\perp} (ax)$ (in μ), and $\frac{}{\vdash X_2^\perp, X_5^\perp} (ax)$ (in ϕ), there is an ax -rule $\frac{}{\vdash X_3^\perp, X_5^\perp} (ax)$ in any normal form of this proof, hence also in τ by Item (c) and Lemma 4 – see the upper half of Figure 2.

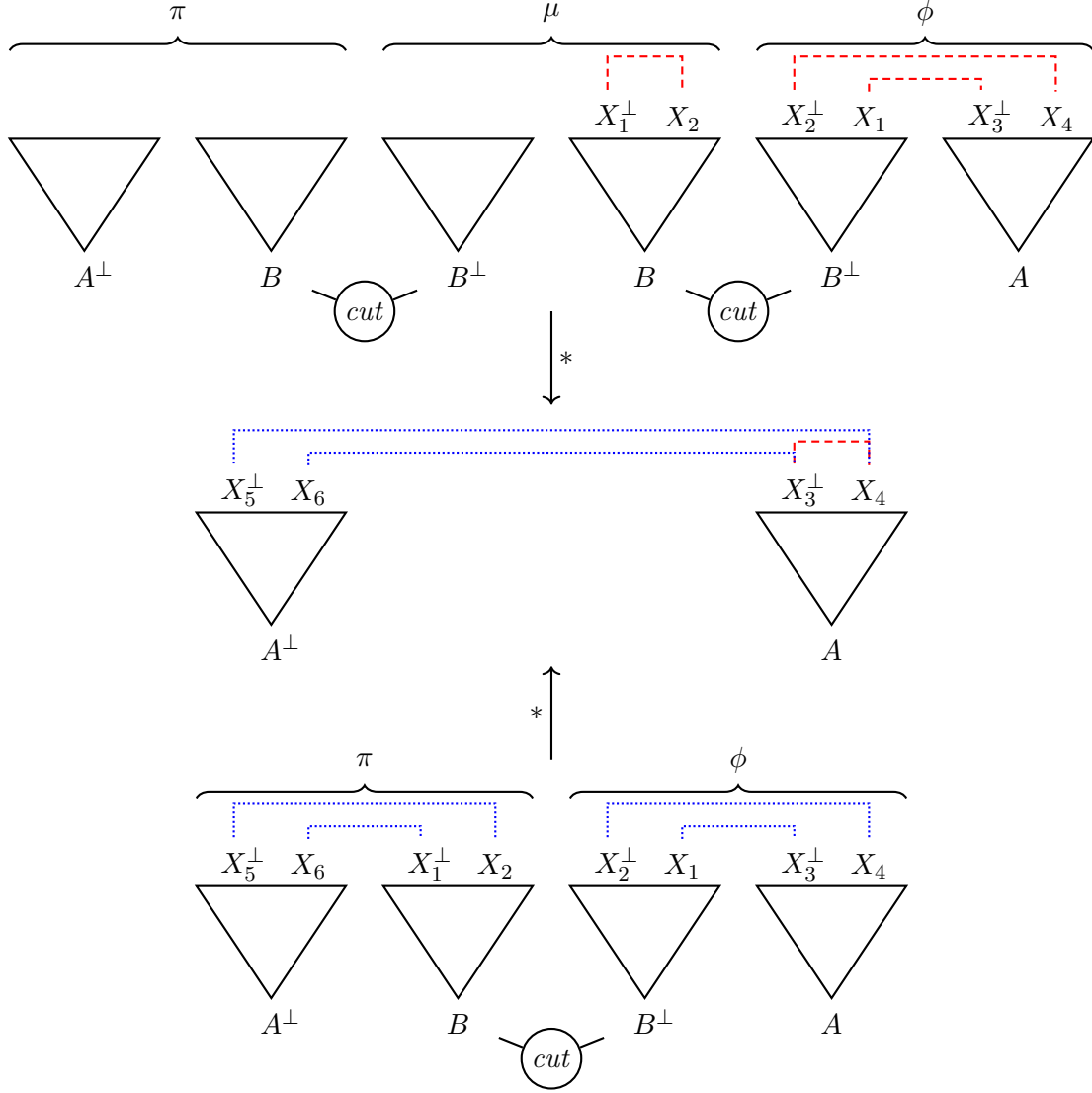


Figure 1: Illustration of the *ax*-rules found in the proof of Lemma 5

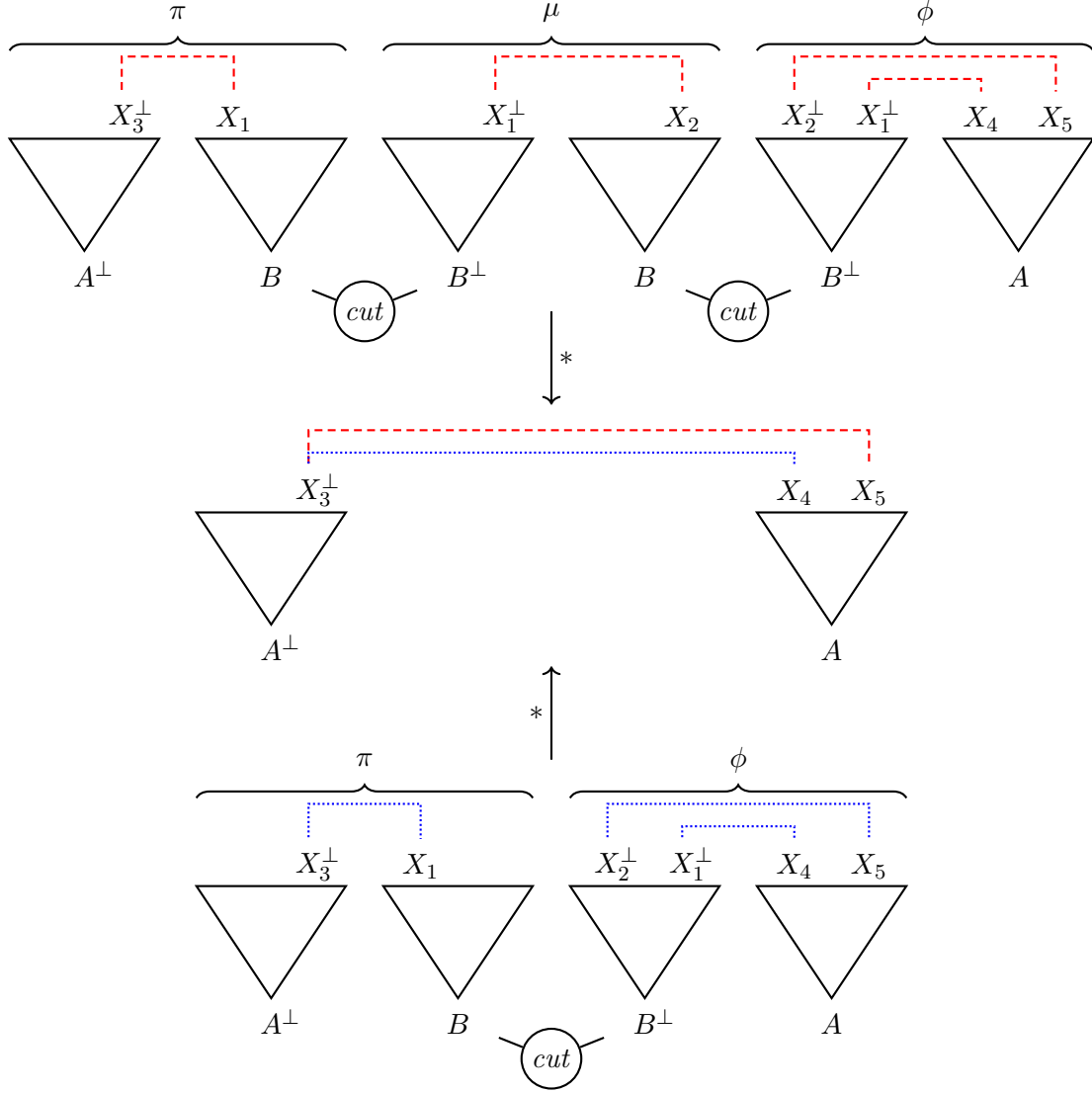


Figure 2: Illustration of the *ax*-rules found in the proof of Lemma 6

Now, let us look at $\frac{\pi \quad \phi}{\vdash A^\perp, B \quad \vdash B^\perp, A} (cut)$, that reduces to τ by Item (a). Thanks to Fact 3 and following the chain of ax -rules $\frac{}{\vdash X_3^\perp, X_1} (ax)$ (in π), and $\frac{}{\vdash X_1^\perp, X_4} (ax)$ (in ϕ), there is an ax -rule $\frac{}{\vdash X_3^\perp, X_4} (ax)$ in τ – see the lower half of Figure 2.

To sum up, we have $\frac{}{\vdash X_3^\perp, X_5} (ax)$ and $\frac{}{\vdash X_3^\perp, X_4} (ax)$ in τ . Since a unique ax -rule contains the occurrence $X_3^\perp$ (Fact 2), X_5 is the same occurrence as X_4 . But then, using $\frac{}{\vdash X_1^\perp, X_4} (ax)$ and $\frac{}{\vdash X_2^\perp, X_5} (ax)$ in ϕ with the same argument, it follows that $X_1^\perp$ and $X_2^\perp$ are the same occurrence. This concludes the proof. $\square$

Putting everything together, we obtain our main result.

Theorem 7. *Unit-free multiplicative linear logic is idempotent complete.*

Explicitly, given an idempotent τ of conclusion $\vdash A^\perp, A$, there exists a section-

retraction pair (π, ϕ) such that $\tau =_{\beta\eta} \frac{\pi \quad \phi}{\vdash A^\perp, B \quad \vdash B^\perp, A} (cut)$.

Proof. Without any loss of generality, we assume that τ is cut -free and ax -expanded. Were it not the case, we consider a normal form τ' of τ instead, using that $\tau' =_{\beta\eta} \tau$.

Theorem 1 applied on τ yields a formula B and two proofs π and ϕ respectively of

$\vdash A^\perp, B$ and of $\vdash B^\perp, A$ such that $\frac{\pi \quad \phi}{\vdash A^\perp, B \quad \vdash B^\perp, A} (cut) =_{\beta\eta} \tau$ and no ax -rule in π (respectively ϕ) uses two occurrences from B (respectively from $B^\perp$). It follows from

Lemmas 4, 5 and 6 that $\frac{\phi \quad \pi}{\vdash B^\perp, A \quad \vdash A^\perp, B} (cut) =_{\beta\eta} \frac{}{\vdash B^\perp, B} (ax)$ since the ax -rules of

the full axiom-expansion of $\frac{}{\vdash B^\perp, B} (ax)$ are exactly between dual occurrences from $B^\perp$ and B (easily provable by induction over B). So (π, ϕ) is a section-retraction pair. $\square$

5 Conclusion

Given a section-retraction pair (π, ϕ) , the proof $\frac{\pi \quad \phi}{\vdash A^\perp, B \quad \vdash B^\perp, A} (cut)$ is always an idempotent. We prove the reciprocal for unit-free multiplicative linear logic: all idempotents are of this shape, *a.k.a.* are split idempotents. This means that from section-retraction pairs one can obtain all idempotents, and conversely.

It would be interesting to check whether this result extends to bigger fragments of linear logic. Notably, one may add units, additive connectives, exponential connectives

or quantifiers. We think that adding the additive connectives $\&$ and $\oplus$ is easy: it should suffice to work slice by slice [Gir87; Gir96], proceeding in each slice as in the unit-free multiplicative fragment. The only delicate point is explaining that slices of the expanded ax -rule on $\vdash B^\perp$, B are in bijection with slices *that are preserved during cut-elimination* of the composition by cut of π and ϕ with as cut -formula B . First-order quantifiers also may be quite easy to add. On the other hand, units seems hard to add, except in presence of the *mix*-rules in which case the proof of this paper trivially extend.

Acknowledgments

Thanks to Vincent Moreau for pointing out this interesting problem to the author.

References

- [BN93] Marcel Bökstedt and Amnon Neeman. “Homotopy limits in triangulated categories”. en. In: *Compositio Mathematica* 86.2 (1993), pp. 209–234. URL: https://www.numdam.org/item/CM_1993__86_2_209_0/.
- [Di +26] Rémi Di Guardia, Thomas Ehrhard, Jérôme Evrard, and Claudia Faggian. “Bayesian Networks and Proof-Nets: the proof-theory of Bayesian Inference”. 2026. URL: <https://arxiv.org/abs/2602.04045>.
- [Di 24] Rémi Di Guardia. “Identity of Proofs and Formulas using Proof-Nets in Multiplicative-Additive Linear Logic”. Thèse de Doctorat. Ecole normale supérieure de lyon - ENS LYON, Sept. 2024. URL: <https://theses.hal.science/tel-04830060>.
- [FOS25] Guido Fiorillo, Daniel Osorio Valencia, and Alexis Saurin. “On Correctness, Sequentialization and Interpolation”. In: *Ninth International Workshop on Trends in Linear Logic and Applications 2025*. Ed. by Lionel Vaux Auclair. 2025. URL: <https://lipn.univ-paris13.fr/TLLA/2025/abstracts/14-fiorillo-osoriovalencia-saurin.pdf>.
- [Gir87] Jean-Yves Girard. “Linear logic”. In: *Theoretical Computer Science* 50 (1987), pp. 1–102. DOI: [10.1016/0304-3975\(87\)90045-4](https://doi.org/10.1016/0304-3975(87)90045-4).
- [Gir96] Jean-Yves Girard. “Proof-nets: the parallel syntax for proof-theory”. In: *Logic and Algebra*. Ed. by Aldo Ursini and Paolo Agliano. Vol. 180. Lecture Notes In Pure and Applied Mathematics. New York: Marcel Dekker, 1996, pp. 97–124. DOI: [10.1201/9780203748671-4](https://doi.org/10.1201/9780203748671-4).
- [HG16] Dominic Hughes and Rob van Glabbeek. *MALL proof nets identify proofs modulo rule commutation*. 2016. URL: <https://arxiv.org/abs/1609.04693>.
- [LL 22] The LL Handbook Project. *Handbook of Linear Logic (draft)*. Early draft for the Linear Logic Winter School. Jan. 2022. URL: <https://ll-handbook.frama.io/ll-handbook/ll-handbook-public.pdf>.

- [Sau25] Alexis Saurin. “Interpolation as Cut-Introduction: On the Computational Content of Craig-Lyndon Interpolation”. In: *International Conference on Formal Structures for Computation and Deduction (FSCD)*. Ed. by Maribel Fernández. Vol. 337. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl – Leibniz-Zentrum für Informatik, July 2025, 32:1–32:21. DOI: [10.4230/LIPIcs.FSCD.2025.32](https://doi.org/10.4230/LIPIcs.FSCD.2025.32).
- [Sch11] Olaf M. Schnürer. *Homotopy categories and idempotent completeness, weight structures and weight complex functors*. 2011. URL: <https://arxiv.org/abs/1107.1227>.
- [See89] Robert Seely. “Linear logic, $\star$ -autonomous categories and cofree coalgebras”. In: *Contemporary mathematics* 92 (1989). URL: <http://www.math.mcgill.ca/rags/nets/lisac.ps.gz>.
- [Shu17] Michael Shulman. “Idempotents in intensional type theory”. In: *Logical Methods in Computer Science* Volume 12, Issue 3, 9 (Apr. 2017). ISSN: 1860-5974. DOI: [10.2168/LMCS-12\(3:9\)2016](https://doi.org/10.2168/LMCS-12(3:9)2016). URL: <https://lmcs.episciences.org/2027>.